

The Ecological Theater

Core theory for the comprehensive exam in ecology and wildlife science: a companion to the narrated program.

42:15 running time **9** chapters **128** lexicon terms **104** references

<i>I</i>	Population Dynamics	0:43	<i>II</i>	Niche Theory	7:10
<i>III</i>	Competition	12:04	<i>IV</i>	Predation	18:13
<i>V</i>	Island Biogeography	24:23	<i>VI</i>	Metapopulations	29:01
<i>VII</i>	Metacommunities	33:06	<i>VIII</i>	Neutral Theory	36:26
<i>IX</i>	Synthesis	40:08			

Each chapter lists the key equations, the landmark studies an examiner is likely to cite, every numbered lexicon term from the video (with the time it appears), and practice prompts. Work the prompts aloud and sketch each graph from memory: isoclines, rate curves, phase planes. Timestamps match the video and the interactive player.

I Population Dynamics

How, and why, do populations change in size?

video 0:43–7:10

0:51 Exponential & geometric growth · 1:58 Logistic growth & density dependence · 3:05 Time lags, cycles & chaos · 4:08 Life tables & matrix models · 5:33 Stochasticity, viability & life history

KEY EQUATIONS

Exponential growth

$$dN/dt = rN, \quad N(t) = N_0 e^{rt}, \quad r = b - d, \quad t_d = \ln 2 / r$$

Geometric growth

$$N_{t+1} = \lambda N_t, \quad \lambda = e^r$$

Logistic growth

$$dN/dt = rN(1 - N/K); \quad \text{MSY} = rK/4 \text{ at } N = K/2$$

Delayed logistic (Hutchinson)

$$dN/dt = rN(t)[1 - N(t - \tau)/K]; \quad \text{monotonic if } r\tau < 1/e; \text{ damped if } 1/e < r\tau < \pi/2; \text{ limit cycle if } r\tau > \pi/2$$

Logistic map (May)

$$x_{t+1} = rx_t(1 - x_t); \quad \text{2-cycle at } r = 3, \text{ 4-cycle at } \approx 3.449, \text{ chaos beyond } \approx 3.57$$

Life table

$$R_0 = \sum l_x m_x; \quad G = \sum x l_x m_x / R_0; \quad r \approx \ln R_0 / G$$

Matrix model

$$\mathbf{n}(t+1) = \mathbf{A} \mathbf{n}(t); \quad \lambda = \text{dominant eigenvalue}; \quad \mathbf{w} = \text{stable stage distribution}; \quad \mathbf{v} = \text{reproductive values}; \quad \text{sensitivity } s_{ij} = v_i w_j / \langle \mathbf{v}, \mathbf{w} \rangle; \quad \text{elasticity } e_{ij} = (a_{ij} / \lambda) s_{ij}$$

Variable environments

$$\lambda_G = (\prod \lambda_t)^{1/T} \leq \bar{\lambda}; \quad \ln \lambda_G \approx \ln \bar{\lambda} - \sigma^2 / (2\bar{\lambda}^2)$$

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Hutchinson 1948	Theory	Time lags in density dependence can generate oscillations.
May 1976	Logistic map	Simple deterministic models produce period doubling and chaos.

STUDY	SYSTEM	KEY FINDING
Crouse, Crowder & Caswell 1987	Loggerhead sea turtle	λ most elastic to large-juvenile survival; supported turtle excluder devices over egg protection alone.
Shaffer 1981	Minimum viable population	Defined MVP via a persistence probability and time horizon (e.g., 99% for 1000 years).
Gilpin & Soulé 1986	Extinction vortex	Genetic and demographic processes reinforce declines in small populations.
Grime 1977	Plant strategies	C-S-R triangle of competitors, stress-tolerators, and ruderals.

LEXICON

No. 001 0:52
Intrinsic rate of increase (r)
 Instantaneous per-capita growth rate, $b - d$, under ideal conditions; units of 1/time.

No. 003 1:41
Density independence
 Per-capita birth and death rates that do not change with population density.

No. 005 2:15
Carrying capacity (K)
 Equilibrium density at which per-capita births equal deaths.

No. 007 2:29
Compensation & overcompensation
 Recruitment that saturates (Beverton–Holt) versus recruitment that declines at high density (Ricker).

No. 009 3:06
Time lag
 Delay between a change in density and the resulting change in per-capita growth.

No. 011 3:34
Period-doubling bifurcation
 Abrupt doubling of a cycle's period as a parameter increases: $1 \rightarrow 2 \rightarrow 4 \rightarrow \dots \rightarrow \text{chaos}$.

No. 002 1:19
Finite rate of increase (λ)
 Ratio of population sizes in successive time steps, $N(t+1)/N(t)$; $\lambda = e^r$.

No. 004 1:59
Density dependence
 Per-capita birth or death rates that change with population density; the basis of population regulation.

No. 006 2:15
Maximum sustainable yield
 Largest harvest a population can sustain indefinitely; at $N = K/2$ in the logistic model.

No. 008 2:47
Allee effect
 Positive density dependence at low density; a strong Allee effect produces an extinction threshold.

No. 010 3:16
Limit cycle
 A self-sustaining oscillation of fixed amplitude that trajectories approach from any start.

No. 012 3:53
Deterministic chaos
 Bounded, aperiodic dynamics from a deterministic rule, with sensitive dependence on initial conditions.

No. 013 4:09
Survivorship (l_x)
Proportion of a cohort surviving from birth to age x .

No. 015 4:54
Leslie & Lefkovitch matrices
Projection matrices of age- or stage-specific survival and fecundity; dominant eigenvalue = λ .

No. 017 5:13
Elasticity
Proportional change in λ per proportional change in a vital rate; elasticities sum to 1.

No. 019 5:34
Environmental stochasticity
Temporal variation in vital rates that affects all individuals together; catastrophes are extreme cases.

No. 021 6:12
PVA & minimum viable population
Projection of extinction risk; MVP = smallest population with an acceptable persistence probability.

No. 023 6:45
r- and K-selection
Life-history strategies favored at low density and high variability (r) versus near carrying capacity (K).

No. 014 4:41
Net reproductive rate (R_0)
 $R_0 = \sum l_x m_x$: expected lifetime offspring per individual; growth per generation.

No. 016 4:54
Stable stage distribution
Constant proportions among stages approached under fixed vital rates (right eigenvector).

No. 018 5:34
Demographic stochasticity
Chance variation in individual births and deaths; strongest in small populations.

No. 020 5:55
Geometric mean growth
Long-run λ under variability; always $\leq$ the arithmetic mean, so variance reduces growth.

No. 022 6:32
Extinction vortex
Mutually reinforcing genetic and demographic declines in small populations (Gilpin & Soulé 1986).

EXAM PROMPTS

1. Derive maximum sustainable yield from the logistic model. Why is harvesting at MSY risky when the environment varies?
2. Show with a numerical example why variance in λ lowers long-run growth even when mean λ exceeds 1.
3. Distinguish sensitivity from elasticity. How did elasticity analysis change sea turtle conservation?
4. How does a strong Allee effect change the logic of minimum viable population size?

II Niche Theory

video 7:10–12:04

Where can a species live, and what does it do there?

7:18 Three ideas of the niche · 8:23 Fundamental & realized niches · 9:23 Partitioning & limiting similarity · 10:50 The modern niche

KEY EQUATIONS

MacArthur–Levins overlap $\alpha_{ij} = \sum_k p_{ik} p_{jk} / \sum_k p_{ik}^2$ (p = proportional use of resource k)

Pianka overlap $O_{jk} = \sum p_{ij} p_{ik} / \sqrt{(\sum p_{ij}^2 \sum p_{ik}^2)}$

Limiting similarity coexistence requires roughly $d/w \gtrsim 1$ (separation of means / utilization width)

Requirement niche (ZNGI) the set of resource levels where $(1/N) dN/dt = f(R_1, R_2) - m = 0$

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Grinnell 1917	California thrasher	Niche as the habitat conditions a species requires.
Elton 1927	Animal Ecology	Niche as functional role: relations to food and enemies.
Hutchinson 1957	Theory	Niche as an n-dimensional hypervolume; fundamental vs realized niches.
Connell 1961	Barnacles, Isle of Cumbrae	Competition from <i>Balanus</i> sets the lower limit of <i>Chthamalus</i> ; desiccation sets <i>Balanus</i> 's upper limit.
MacArthur 1958	Spruce-woods warblers	Five congeners partition foraging zones and behavior within trees.
Lack 1947; Grant & Grant 2006	Darwin's finches	Character displacement in beak size in sympatry.
Chase & Leibold 2003	Theory	Niche = requirements (ZNGIs) + impacts (consumption vectors).

STUDY	SYSTEM	KEY FINDING
Soberón & Peterson 2005	Distribution modeling	BAM diagram: Abiotic, Biotic, and Movement constraints on distributions.

LEXICON

No. 024 7:19	No. 025 7:36
Grinnellian niche The habitat and environmental conditions a species requires; the basis of distribution modeling.	Eltonian niche A species' functional role: what it eats, what eats it, and its effects on the community.
No. 026 7:52	No. 027 8:24
Hutchinsonian niche The n-dimensional hypervolume of conditions and resources permitting a species to persist.	Fundamental niche Conditions under which a species can persist given abiotic tolerances alone, without enemies or competitors.
No. 028 8:24	No. 029 9:24
Realized niche The portion of the fundamental niche a species actually occupies, given biotic interactions.	Niche partitioning Division of resources, space, or time among coexisting species, reducing competition.
No. 030 9:42	No. 031 9:58
Niche overlap Shared use of resources by two species, measured as overlap of utilization curves.	Limiting similarity The maximum similarity in resource use compatible with stable coexistence (MacArthur & Levins 1967).
No. 032 10:18	No. 033 10:37
Character displacement Evolutionary divergence of traits where competing species co-occur (sympatry) but not apart (allopatry).	Ghost of competition past Connell (1980): present niche differences produced by past competition, now no longer observable.
No. 034 10:51	No. 035 11:07
Requirement & impact niches Chase & Leibold (2003): conditions allowing zero or positive growth, and the species' per-capita effects on them.	ZNGI Zero net growth isocline: resource levels at which a population's growth exactly balances its losses.
No. 036 11:30	No. 037 11:30
Species distribution model Statistical model relating occurrences to environmental predictors to map habitat suitability.	BAM diagram Soberón & Peterson: occupied area = overlap of Abiotic suitability, Biotic interactions, and Movement.

Niche construction

Modification of the environment by organisms, altering selection on themselves and others.

EXAM PROMPTS

1. Compare Grinnellian, Eltonian, and Hutchinsonian niches. Which does a species distribution model estimate, and why?
2. Design an experiment that separates the fundamental from the realized niche of two plants along a moisture gradient.
3. What evidence would distinguish character displacement from alternative explanations (cf. Schluter & McPhail 1992)?
4. Explain how requirement and impact niches together predict the outcome of competition.

III Competition

video 12:04–18:13

What happens when species need the same things?

12:12 Kinds of competition · 13:52 Lotka–Volterra competition · 15:25 Resource competition & R^* · 16:36 Modern coexistence theory

KEY EQUATIONS

Lotka–Volterra competition	$\frac{dN_1}{dt} = r_1 N_1 (K_1 - N_1 - \alpha_{12} N_2) / K_1; \quad \frac{dN_2}{dt} = r_2 N_2 (K_2 - N_2 - \alpha_{21} N_1) / K_2$
Outcomes	species 1 wins: $K_1 > K_2 / \alpha_{21}$ and $K_1 / \alpha_{12} > K_2$ · species 2 wins: reverse · stable coexistence: $K_1 < K_2 / \alpha_{21}$ and $K_2 < K_1 / \alpha_{12}$ ($\alpha_{12} < K_1 / K_2$, $\alpha_{21} < K_2 / K_1$) · unstable (priority effects): $K_1 > K_2 / \alpha_{21}$ and $K_2 > K_1 / \alpha_{12}$
Resource competition (Tilman)	$\frac{dN}{dt} = N[f(R) - m]; \quad R^* \text{ solves } f(R^*) = m; \text{ with Monod uptake } f = \mu_{\max} R / (K_s + R), \quad R^* = m K_s / (\mu_{\max} - m)$
Modern coexistence (Chesson)	coexistence when $\rho < \kappa_1 / \kappa_2 < 1 / \rho$; niche difference = $1 - \rho$; fitness ratio = κ_1 / κ_2 ; invasion criterion: $\bar{r}_i > 0$ when rare

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Gause 1934	Paramecium	Competitive exclusion in laboratory cultures.
Park 1954	Tribolium flour beetles	Climate-dependent outcomes and priority effects.
Tilman 1977	Freshwater diatoms	Phosphate and silicate ratios predicted competitive outcomes (R^* and resource ratios).
Wedin & Tilman 1993	Prairie grasses	Lowest R^* for soil nitrogen predicted the winner.
Schoener 1983	Review of field experiments	Competition common in field experiments; six mechanisms of competition.
Holt 1977	Theory	Apparent competition through a shared predator.

STUDY	SYSTEM	KEY FINDING
Janzen 1970; Connell 1971	Tropical trees	Specialist enemies near parents promote diversity (Janzen–Connell).
Chesson 2000	Theory	Stabilizing vs equalizing mechanisms; storage effect; relative nonlinearity.
Hutchinson 1961	Phytoplankton	Paradox of the plankton.

LEXICON

No. 039 12:13 Intra- vs interspecific competition Competition within a species (the source of density dependence) versus between species.	No. 040 12:30 Exploitative vs interference Indirect competition through resource depletion versus direct prevention of resource use.
No. 041 12:48 Allelopathy Chemical interference: release of compounds that inhibit competitors (e.g., juglone of black walnut).	No. 042 13:08 Scramble vs contest Even division of resources so all suffer, versus winners taking enough and losers none (Nicholson 1954).
No. 043 13:08 Asymmetric competition Unequal per-capita effects between competitors; e.g., size-asymmetric competition for light.	No. 044 13:34 Apparent competition Indirect negative interaction between prey species mediated by a shared predator (Holt 1977).
No. 045 13:53 Competition coefficient (α) Per-capita effect of one species on another's growth, relative to intraspecific effects.	No. 046 14:13 Zero-growth isocline All combinations of densities at which a species' growth rate is zero.
No. 047 14:28 Competitive exclusion principle Gause: complete competitors cannot coexist indefinitely; one excludes the other.	No. 048 14:50 Priority effect When arrival order or initial abundance determines the outcome of interactions.
No. 049 15:26 R* Minimum resource level at which a population can persist; with one limiting resource, lowest R* wins.	No. 050 16:19 Supply point Resource availability in the absence of consumers; locates the outcome on the two-resource plane.
No. 051 16:19 Resource-ratio hypothesis Tilman: the ratio of limiting resources supplied determines which species dominate or coexist.	No. 052 16:37 Mutual invasibility Coexistence criterion: each species has a positive growth rate when rare in the other's presence.

No. 053 16:52
Stabilizing mechanism
Niche difference causing stronger intra- than interspecific limitation; favors species when rare.

No. 055 17:27
Janzen–Connell effect
Specialist enemies concentrated near parent trees reduce recruitment of conspecifics nearby.

No. 057 17:53
Paradox of the plankton
Hutchinson (1961): many phytoplankton coexist on few resources, defying competitive exclusion.

No. 054 16:52
Equalizing mechanism
Reduction in average fitness differences between species; cannot by itself stabilize coexistence.

No. 056 17:27
Storage effect
Fluctuation-dependent coexistence: species-specific responses to environment plus buffered population growth.

EXAM PROMPTS

1. Draw the four Lotka–Volterra isocline configurations and state the condition for each outcome.
2. Why does the species with the lowest R^* win? Which assumptions does the rule require?
3. Distinguish stabilizing from equalizing mechanisms, with a field example of each.
4. How would you test for apparent competition in a field system?

IV Predation

video 18:13–24:23

How do consumers and their victims shape each other?

18:19 Predator–prey cycles · 19:32 Functional & numerical responses · 20:49 Optimal foraging · 21:39 Enrichment & stability · 22:49 Keystones, cascades & fear

KEY EQUATIONS

Lotka–Volterra predator–prey $dN/dt = rN - aNP$; $dP/dt = caNP - mP$; isoclines $N^* = m/(ca)$, $P^* = r/a$

Functional responses (Holling) Type II: $f(N) = aN/(1 + ahN) \rightarrow 1/h$ as $N \rightarrow \infty$; Type III: $f(N) = aN^2/(1 + ahN^2)$

Rosenzweig–MacArthur $dN/dt = rN(1 - N/K) - aNP/(1 + ahN)$; $dP/dt = caNP/(1 + ahN) - mP$

Nicholson–Bailey $H_{t+1} = \lambda H_t e^{-aP_t}$; $P_{t+1} = cH_t(1 - e^{-aP_t})$

Marginal value theorem leave when $g'(t^*) = g(t^*)/(\tau + t^*)$, the long-term average intake rate including travel time τ

Diet model add prey type i (ranked by E/h) if $E_i/h_i > \Sigma_{j<i} \lambda_j E_j / (1 + \Sigma_{j<i} \lambda_j h_j)$

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Gause 1934	Paramecium & Didinium	Predator eliminated prey unless refuges or immigration were added.
Huffaker 1958	Mites on oranges	Persistence required spatial complexity and dispersal barriers.
Holling 1959	Functional responses	Types I–III; the disc equation and handling time.
Rosenzweig 1971	Theory	Paradox of enrichment.

STUDY	SYSTEM	KEY FINDING
Krebs et al. 1995; Sheriff et al. 2009	Snowshoe hare, Kluane (Yukon)	Food and predation jointly drive the cycle; predator-induced stress lowers reproduction.
Paine 1966, 1969	Pisaster, Washington coast	Keystone predation maintains diversity (15 → 8 species after removal).
Hairston, Smith & Slobodkin 1960	Theory	Green world hypothesis: predators limit herbivores.
Estes et al. 1998	Aleutian kelp forests	Orca predation on otters released urchins and collapsed kelp.
Crooks & Soulé 1999	Southern California scrub	Coyote loss led to mesopredator release and bird declines.
Zanette et al. 2011	Song sparrows	Predator playback alone reduced annual offspring by ~40%.
Charnov 1976	Theory	Marginal value theorem for patch residence.

LEXICON

No. 058 18:20 Lotka–Volterra predator–prey model $dN/dt = rN - aNP$; $dP/dt = caNP - mP$: exponential prey, linear functional response.	No. 059 18:53 Neutral cycles Closed orbits whose amplitude depends only on initial conditions; structurally unstable.
No. 060 19:11 Population cycle Regular multi-year oscillation in abundance, e.g., the ~10-year snowshoe hare–lynx cycle.	No. 061 19:33 Functional response Per-predator consumption rate as a function of prey density (Holling Types I, II, III).
No. 062 19:43 Handling time (h) Time to pursue, subdue, and consume one prey item; caps intake at $1/h$ in a Type II response.	No. 063 20:04 Prey switching Disproportionate attack on whichever prey type is most common; one cause of Type III responses.
No. 064 20:33 Numerical response Change in predator density with prey density, via reproduction or aggregation.	No. 065 20:33 Total response Functional × numerical response: total prey killed per unit time.

No. 066	20:50	No. 067	21:00
Optimal foraging theory		Profitability	
Predicts foraging decisions that maximize net energy intake rate.		Energy gained per unit handling time (E/h) for a prey type; the basis of diet ranking.	
No. 068	21:19	No. 069	21:40
Marginal value theorem		Rosenzweig–MacArthur model	
Charnov (1976): leave a patch when its marginal intake rate falls to the habitat-wide average.		Predator–prey model with logistic prey and a Type II functional response; hump-shaped prey isocline.	
No. 070	22:08	No. 071	22:26
Paradox of enrichment		Nicholson–Bailey model	
Rosenzweig (1971): raising prey carrying capacity destabilizes predator–prey dynamics.		Discrete host–parasitoid model with diverging oscillations; stabilized by aggregation or refuges.	
No. 072	22:26	No. 073	22:50
Prey refuge		Keystone species	
Habitat or condition protecting part of the prey population, which tends to stabilize dynamics.		A species whose impact on its community is disproportionately large relative to its abundance (Paine).	
No. 074	23:33	No. 075	23:33
Trophic cascade		Top–down vs bottom–up	
Alternating indirect effects of predators down a food chain, e.g., otters → urchins → kelp.		Regulation of populations by consumers above them versus by resources below them.	
No. 076	23:33	No. 077	23:55
Mesopredator release		Non–consumptive effects	
Increase in smaller predators after loss of an apex predator (Crooks & Soulé 1999).		Effects of predation risk on prey behavior, physiology, and demography without killing.	
No. 078	23:55		
Landscape of fear			
Spatial variation in prey-perceived predation risk that shapes habitat use.			

EXAM PROMPTS

1. Why is the Lotka–Volterra predator–prey model structurally unstable? List modifications that stabilize it.
2. Explain the paradox of enrichment with the Rosenzweig–MacArthur isocline diagram.
3. Contrast consumptive and non-consumptive effects of predators. How could you separate them experimentally?
4. Under what conditions should a trophic cascade propagate to primary producers?

V Island Biogeography

Why do islands hold the species they do?

video 24:23–29:01

24:32 The species–area relationship · 25:32 The equilibrium theory · 26:47 Experiments & refinements · 27:55 Habitat islands & conservation

KEY EQUATIONS

Species–area relationship

$S = cA^z$; $\log S = \log c + z \log A$; islands $z \approx 0.20\text{--}0.35$, mainland nested samples $\approx 0.12\text{--}0.17$

Linear equilibrium model

$I(S) = I_0(1 - S/P)$, $E(S) = E_0S/P \Rightarrow \hat{S} = I_0P/(I_0 + E_0)$, turnover $T = I_0E_0/(I_0 + E_0)$

Darlington's rule

with $z \approx 0.3$, a tenfold loss of area leaves $10^{-0.3} \approx 0.5$ of the species

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
MacArthur & Wilson 1963, 1967	Theory	Equilibrium theory: richness balances immigration and extinction.
Simberloff & Wilson 1969	Mangrove islets, Florida Keys	Arthropods recolonized to near pre-defaunation richness within ~1 year, with turnover.
Brown & Kodric-Brown 1977	Arthropods on thistles	Rescue effect: immigration lowers extinction on near islands.
Lomolino 1990	Mammals on islands	Target-area effect: larger islands receive more immigrants.
Diamond 1975; Simberloff & Abele 1976	Reserve design	The SLOSS debate.
Newmark 1987	Western North American parks	Mammal extinctions since establishment, greatest in small parks.
Whittaker et al. 2008	Oceanic islands	General dynamic model: island age, area, and in-situ speciation.

LEXICON

No. 079 24:33
Species–area relationship
S = cA^z: species richness increases as a power function of area.

No. 081 25:33
Equilibrium theory of island biogeography
MacArthur & Wilson: island richness is a dynamic balance of immigration and extinction.

No. 083 26:12
Turnover
Replacement of species at equilibrium: extinction of residents balanced by new arrivals.

No. 085 26:48
Defaunation experiment
Simberloff & Wilson (1969): removing island arthropods to watch recolonization reach equilibrium.

No. 087 27:11
Target effect
Larger islands intercept more dispersers, raising immigration rates.

No. 089 27:56
Habitat fragmentation
Breaking of continuous habitat into smaller, isolated patches within a matrix of altered land.

No. 091 28:16
Extinction debt
Future extinctions already committed by past habitat loss or isolation.

No. 080 24:59
z-value
Slope of log S vs. log A; ~0.2–0.35 for islands, ~0.12–0.17 for mainland samples.

No. 082 25:50
Source pool (P)
The set of species on the mainland available to colonize an island.

No. 084 26:25
Distance & area effects
Isolation lowers immigration; small area raises extinction.

No. 086 27:11
Rescue effect
Immigration that bolsters small populations and lowers extinction rates (Brown & Kodric-Brown 1977).

No. 088 27:36
In situ speciation
On remote islands, new species arise within the archipelago, adding to richness beyond immigration.

No. 090 27:56
Relaxation
Decline of species richness in a newly isolated fragment toward a lower equilibrium (faunal collapse).

No. 092 28:34
SLOSS debate
Single large or several small reserves of equal total area: which conserves more species?

EXAM PROMPTS

1. Draw the MacArthur–Wilson diagram for near/far and large/small islands; predict richness and turnover for each.
2. What did the Simberloff–Wilson defaunation experiment demonstrate, and what were its limitations?
3. Define extinction debt and explain how you would estimate it for a fragmented landscape.
4. Argue both sides of SLOSS for a reserve network in your own study system.

VI Metapopulations

video 29:01–33:06

How does a species persist when every local population winks out?

29:09 The Levins model · 30:17 Habitat loss & the extinction threshold · 31:12 Kinds of spatial populations · 32:17 Spatially realistic metapopulations

KEY EQUATIONS

Levins model

$$dp/dt = cp(1 - p) - ep; \quad p^* = 1 - e/c; \text{ persistence requires } c > e$$

Habitat destruction

$$dp/dt = cp(1 - D - p) - ep; \quad p^* = 1 - D - e/c; \text{ extinction threshold at } D = 1 - e/c$$

Incidence function model (Hanski)

$$J_i = C_i / (C_i + E_i - C_i E_i); \quad E_i = \min(1, e/A_i^x); \quad C_i = S_i^2 / (S_i^2 + y^2)$$

Connectivity

$$S_i = \sum_{j \neq i} p_j e^{-\alpha d_{ij}} A_j^b$$

Source–sink

sources: $\lambda > 1$ (net exporters); sinks: $\lambda < 1$ (persist only with immigration)

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Levins 1969	Agricultural pests	Metapopulation concept and patch-occupancy model.
Harrison 1991	Review	Types: classic, mainland–island, patchy, nonequilibrium.
Pulliam 1988	Theory	Sources, sinks, and population regulation.
Hanski 1994, 1998	Glanville fritillary, Åland	Incidence function model; occupancy predicted by area and connectivity.
Moran 1953	Canadian lynx	Correlated weather synchronizes separate populations (Moran effect).
Tilman, May, Lehman & Nowak 1994	Theory	Habitat destruction and extinction debt; best competitors lost first.

STUDY	SYSTEM	KEY FINDING
Schlaepfer, Runge & Sherman 2002	Review	Ecological and evolutionary traps.

LEXICON

No. 093 29:10 Metapopulation A set of local populations in discrete patches linked by dispersal, with local extinction and recolonization.	No. 094 29:28 Patch occupancy (p) Fraction of habitat patches occupied; the state variable of the Levins model.
No. 095 29:47 Equilibrium occupancy (p^*) $p^* = 1 - e/c$ in the Levins model; the metapopulation persists only if $c > e$.	No. 096 30:31 Extinction threshold Level of habitat loss at which a metapopulation collapses although suitable habitat remains: $D = 1 - e/c$.
No. 097 30:46 Unoccupied suitable habitat Empty patches capable of supporting a population; essential to colonization–extinction dynamics.	No. 098 31:13 Mainland–island metapopulation A large persistent population supplies colonists to small, extinction-prone satellite populations.
No. 099 31:13 Patchy population Patches linked by such frequent dispersal that they function as one population.	No. 100 31:38 Source–sink dynamics Pulliam (1988): surplus from habitats with $\lambda > 1$ sustains populations in habitats with $\lambda < 1$.
No. 101 31:56 Ecological trap Low-quality habitat that animals prefer because once-reliable cues have become decoupled from quality.	No. 102 32:18 Incidence function model Hanski (1994): patch occupancy probability as a function of patch area and connectivity.
No. 103 32:18 Connectivity Distance-weighted influence of the surrounding occupied patches, scaled by their areas (Hanski's S_j).	No. 104 32:48 Spatial synchrony Correlated fluctuations among local populations, raising the risk of simultaneous extinction.
No. 105 32:48 Moran effect Synchrony among populations induced by correlated environmental variation such as shared weather.	

EXAM PROMPTS

1. Derive p^* for the Levins model and explain why a species can vanish while suitable habitat remains.
2. Distinguish the main metapopulation types and give a real example of each.
3. Why can abundance mislead about habitat quality? Include ecological traps in your answer.

4. How does spatial synchrony affect metapopulation persistence, and what causes it?

VII Metacommunities

video 33:06–36:26

How do local communities connect across a region?

33:13 Scale & diversity · 34:24 Four paradigms · 35:30 Dispersal & diagnosis

KEY EQUATIONS

Whittaker's partition

$$\gamma = \bar{\alpha} \times \beta \text{ (multiplicative); additive form } \gamma = \bar{\alpha} + \beta_{\text{add}}$$

Turnover vs nestedness
(Baselga)

$$\beta_{\text{SOR}} = \beta_{\text{SIM}} \text{ (turnover)} + \beta_{\text{SNE}} \text{ (nestedness-resultant)}$$

Variation partitioning

$$[a] \text{ environment only} + [b] \text{ shared} + [c] \text{ space only} + [d] \text{ unexplained} \\ = 1 \text{ (adjusted } R^2 \text{ from RDA)}$$

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Whittaker 1960, 1972	Siskiyou Mountains vegetation	Introduced α , β , and γ diversity.
Tilman 1994	Theory	Competition–colonization trade-off permits regional coexistence.
Leibold et al. 2004	Synthesis	Metacommunity concept; four paradigms.
Mouquet & Loreau 2003	Theory	Local diversity peaks at intermediate dispersal; β declines with dispersal.
Cottenie 2005	Meta-analysis of 158 datasets	Variation partitioning across metacommunities; species sorting most common.
Kraft et al. 2015	Critique	Competition can masquerade as environmental filtering.

LEXICON

No. 106

Metacommunity

A set of local communities linked by dispersal of multiple potentially interacting species (Leibold et al. 2004).

33:14

No. 107

Alpha, beta & gamma diversity

Local richness, among-site compositional variation, and regional richness; $\gamma = \alpha \times \beta$ (Whittaker).

33:28

No. 108 33:47
Turnover vs nestedness
Beta diversity from species replacement versus from poorer sites holding subsets of richer ones.

No. 110 34:25
Patch dynamics
Paradigm of identical patches where coexistence arises from a competition–colonization trade-off.

No. 112 34:43
Species sorting
Paradigm in which environmental heterogeneity and niche differences determine local composition.

No. 114 35:31
Dispersal limitation
Failure of species to reach all suitable sites, leaving composition spatially structured.

No. 109 34:01
Environmental filtering
Exclusion of species unable to tolerate local abiotic conditions during community assembly.

No. 111 34:25
Competition–colonization trade-off
Better competitors are poorer colonizers, allowing fugitive species to persist regionally.

No. 113 34:59
Mass effects
Paradigm in which high dispersal sustains species in sink patches, decoupling composition from environment.

No. 115 35:49
Variation partitioning
Dividing explained compositional variation into pure environmental, pure spatial, and shared fractions.

EXAM PROMPTS

1. Contrast the four metacommunity paradigms by their assumptions about environmental heterogeneity and dispersal.
2. Why is local diversity predicted to peak at intermediate dispersal rates?
3. What can, and cannot, variation partitioning tell you about metacommunity processes?
4. Partition γ into α and β for three sites with species {A,B,C}, {B,C,D}, and {D,E}.

VIII Neutral Theory

What if species differences didn't matter at all?

video 36:26–40:08

36:34 Ecological equivalence · 37:46 θ , m & abundance distributions · 39:06 Tests & legacy

KEY EQUATIONS

Fundamental biodiversity number	$\theta = 2J_M\nu$ (metacommunity size $\times$ speciation rate); equals Fisher's α at equilibrium
Log-series (metacommunity)	expected species with n individuals: $S_n = \theta x^n/n$, with x close to 1
Ewens sampling formula	$E[S] = \sum_{k=0}^{J-1} \theta/(\theta + k)$
Immigration	m = probability a local death is replaced by an immigrant; Etienne's $I = m(J - 1)/(1 - m)$

LANDMARK STUDIES

STUDY	SYSTEM	KEY FINDING
Kimura 1968	Molecular evolution	Neutral theory of molecular evolution; drift of neutral alleles.
Fisher, Corbet & Williams 1943	Butterflies, moths	Log-series distribution and Fisher's α .
Preston 1948, 1962	Birds, moths	Lognormal distribution in octaves; veil line.
Hubbell 2001	Barro Colorado Island forest	Unified neutral theory; θ , m , zero-sum multinomial.
Condit et al. 2002	Tropical forests, Panama and Amazonia	Distance decay of similarity tested against dispersal limitation.
McGill 2003; Volkov et al. 2003	BCI and others	Debate over whether neutral SADs outperform the lognormal.
Comita et al. 2010	Barro Colorado Island	Conspecific negative density dependence varies among species.

STUDY	SYSTEM	KEY FINDING
Scheffer & van Nes 2006	Theory	Emergent neutrality: clumps of similar species.

LEXICON

No. 116	36:51	No. 117	37:05
Ecological equivalence		Zero-sum dynamics	
Identical per-capita rates of birth, death, dispersal, and speciation for all individuals of all species.		A community of fixed size J in which each death is matched by one birth or immigrant.	
No. 118	37:05	No. 119	37:29
Ecological drift		Monodominance	
Random change in species' relative abundances due to demographic stochasticity.		Fixation of a single species, the inevitable endpoint of drift in a closed neutral community.	
No. 120	37:47	No. 121	38:13
Fundamental biodiversity number (θ)		Immigration probability (m)	
$\theta = 2 \times \text{metacommunity size} \times \text{speciation rate per birth}$; equals Fisher's α at equilibrium.		Probability that a local death is replaced by an immigrant from the metacommunity.	
No. 122	38:32	No. 123	38:32
Species abundance distribution		Preston's veil line	
Frequency distribution of species' abundances; log-series, lognormal, or zero-sum multinomial forms.		Sampling limit that hides the rarest species, truncating the left side of a lognormal SAD.	
No. 124	38:51	No. 125	39:07
Zero-sum multinomial		Distance decay of similarity	
Hubbell's neutral SAD for a dispersal-limited local community.		Decline in compositional similarity between communities with geographic distance.	
No. 126	39:40	No. 127	39:40
Null model		Emergent neutrality	
A model that excludes the process of interest, providing a baseline for detecting it.		Scheffer & van Nes (2006): clusters of similar species coexisting near-neutrally within niches.	

EXAM PROMPTS

1. List the assumptions of neutral theory. Which are most often violated, and does that matter?
2. What is θ , how is it estimated, and how does it relate to Fisher's α ?
3. Why is fitting a species abundance distribution weak evidence for neutrality?
4. Explain neutral theory's value as a null model, with an example from your field.

IX Synthesis

How do the theories fit together?

video 40:08–42:15

40:14 Four high-level processes · 41:22 Exam rehearsal

FOUR PROCESSES

PROCESS	POPULATION-GENETICS ANALOGUE	WHERE IT APPEARS
Selection	Natural selection	Niches, competition, predation, stabilizing and equalizing mechanisms, species sorting
Ecological drift	Genetic drift	Demographic stochasticity, small populations, neutral theory, island extinction
Dispersal	Gene flow	Island colonization, rescue effect, metapopulations, mass effects
Speciation	Mutation	Point speciation (θ), in-situ radiation on remote islands, regional pool

LEXICON

No. 128

40:15

Vellend's four processes

Selection, ecological drift, dispersal, and speciation: the high-level processes of community dynamics.

REHEARSAL ANSWERS

Why is a Type II functional response destabilizing?

Per-capita predation risk, $f(N)/N = a/(1 + ahN)$, falls as prey density rises, so predation weakens as prey increase: inverse density dependence that amplifies fluctuations.

Why does the species with the lowest R^* win?

At equilibrium each consumer holds the resource at its own R^* . The species with the lower R^* depresses the resource below the other's break-even level, so the competitor's growth becomes negative (single limiting resource, well-mixed, equilibrium conditions).

Why don't empty patches mean habitat is unimportant?

In colonization–extinction dynamics a fraction of suitable patches (e/c in the Levins model) is always empty. Those patches are needed for recolonization; destroying them lowers p^* and can cross the extinction threshold.

Stabilizing vs equalizing mechanisms?

Stabilizing mechanisms (niche differences) make intraspecific limitation exceed interspecific limitation and give rare species an advantage. Equalizing mechanisms shrink average fitness differences. Only stabilizing mechanisms create a rare-species advantage.

How do the four paradigms differ?

Patch dynamics: homogeneous patches, competition–colonization trade-off. Species sorting: heterogeneous patches, moderate dispersal, composition tracks environment. Mass effects: heterogeneous patches, high dispersal, source–sink spillover. Neutral: equivalent species, drift and dispersal limitation.

What does θ measure?

$\theta = 2J_M v$, the product of metacommunity size and speciation rate. It sets regional richness and evenness under neutrality and equals Fisher's α of the log-series.

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